Conflict-Suppression and Self-Organized Criticality

Insights from a Modified Forest-Fire Automaton

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Abstract

We present a stylized cellular automaton model — a modified Drossel–Schwabl forest-fire system — in which a conflict-suppression mechanism is applied to a lattice of geopolitical tension zones. The model is an abstract toy system making no empirical claims about actual peacekeeping operations or geopolitical outcomes. Its purpose is to explore how a tension-suppression operator affects the self-organized critical (SOC) dynamics of a conflict-spreading automaton. A suppression-failure mechanism (p_{fail} = 0.003 per step) allows rare but complete collapses of the quenching pathway, producing catastrophic cascades. We find that increasing suppression strength reduces individual conflict sizes but raises the background tension fraction — the *suppression–loading effect*. The avalanche-size distribution follows an approximate power law $N(s) \sim s^{-\tau}$ with $\tau \approx 1.13$ in the low-suppression regime, broadly consistent with Richardson's empirical war-size statistics.

Keywords: self-organized criticality · forest-fire model · conflict dynamics · cellular automaton · power law · tension accumulation · complexity science

1. Introduction

Self-organized criticality (SOC), introduced by Bak, Tang, and Wiesenfeld (1987, 1988), describes a class of dynamical systems that spontaneously evolve toward a critical state producing scale-free, power-law event-size distributions without external parameter tuning. The canonical examples are the sandpile model and the Drossel–Schwabl forest-fire model (1992).

An intriguing empirical observation, due to Richardson (1948) and revisited by Cederman (2003) and Clauset et al. (2009), is that the frequency–magnitude distribution of armed conflicts approximates a power law over several decades of conflict size, with no characteristic war magnitude. This is qualitatively consistent with SOC behavior.

This project develops and analyzes a stylized toy model in which:

- A lattice of cells represents geopolitical zones with increasing tension states;

- Conflict spreads stochastically to neighboring high-tension zones (the avalanche);
- A suppression operator probabilistically quenches spreading, with a rare failure probability p_{fail} allowing complete collapse of the suppression pathway;
- Tension accumulates slowly via stochastic growth (the noise).

The model is explicitly *not* a realistic simulation of international relations or peacekeeping policy. It is a minimal cellular automaton designed to ask: how does a conflict-suppression mechanism qualitatively affect SOC dynamics? The answer reveals a trade-off between conflict size and background tension loading, culminating in catastrophic cascades when suppression fails on a loaded lattice — the complete forest-fire suppression paradox.

The Forest-Fire Analogy — Explicit Mapping

Forest-Fire Model	This Conflict Model
Fuel accumulation (dry trees)	Geopolitical tension build-up (states 1–5)
Fire spreading to adjacent trees	Conflict cascading to neighbouring zones
Firefighting / suppression crews	Peacekeeping intervention (parameter α)
Rare catastrophic wildfire (suppression paradox)	Catastrophic war cascade on suppression failure

2. Why Is This System a Good Candidate for SOC?

A system is a plausible SOC candidate when it exhibits slow external driving, threshold-mediated dissipation, local interactions, and no characteristic event scale (Jensen, 1998). The stylized conflict model satisfies these conditions as follows.

- **Slow driving / fast dissipation.** Tension accumulates over many time steps (slow driving) but discharges rapidly once ignition occurs (fast dissipation). This timescale separation is the fundamental precondition for SOC.
- **Threshold dynamics.** A cell does not ignite at any tension level; ignition requires either a random spark event or contact with a burning neighbor. This nonlinear threshold is the hallmark of sandpile and forest-fire SOC systems.
- **Local interactions.** Conflict spreads only to von Neumann neighbors (four adjacent cells), creating locally mediated cascades. Global behavior emerges from local rules without any long-range coupling.
- **Absence of a characteristic scale.** In the low-suppression regime, avalanche sizes span several orders of magnitude with no preferred scale, consistent with a power-law distribution.
- **Suppression pathway with failure.** The suppression operator reduces spreading probability under normal operation. The rare failure mechanism (p_{fail} = 0.003) allows complete collapse of the quenching pathway, discharging stored tension in a catastrophic cascade — the novel element that completes the forest-fire analogy.

3. Model Description

3.1 State Space

The model is defined on an N × N lattice (default N = 100). Each cell i carries a state $\sigma_i \in \{0, 1, 2, 3, 4, 5, 6, 7\}$:

State	Code	Interpretation
Empty	0	Post-conflict; no residual tension
Tension levels	1–5	Increasing latent tension
Active conflict	6	Spreading cascade (fire)
Suppressed	7	Conflict quenched by suppression operator

Table 1: Cell state encoding.

3.2 Update Rules

At each discrete step, all cells update synchronously according to six rules:

- **Rule 1 — Suppression failure check.** Draw u ~ Uniform(0,1). If $u < p_{fail} = 0.003$, effective suppression $\alpha_{eff} = 0$ for this step (suppression collapses entirely). Otherwise $\alpha_{eff} = \alpha$.
- **Rule 2 — Conflict spreading.** Each burning cell ($\sigma_i = 6$) attempts to ignite each tension-loaded neighbor ($1 \le \sigma_j \le 5$). With probability α_{eff}, spreading is blocked: $\sigma_j \to 7$. Otherwise conflict spreads: $\sigma_j \to 6$.
- **Rule 3 — Burn-out.** $\sigma_i = 6 \Rightarrow \sigma_i = 0$ (conflict ends, zone resets).
- **Rule 4 — Suppressed zone recovery.** Each suppressed cell ($\sigma_i = 7$) returns to low tension ($\sigma_i = 1$) with probability $p_r = 0.03$ per step.
- **Rule 5 — Tension growth (noise).** $\sigma_i \to \sigma_i + 1$ with probability p = 0.03 if $\sigma_i = 0$, and p/2 if $1 \le \sigma_i \le 4$. Models continuous low-level background perturbations.
- **Rule 6 — Random sparks.** High-tension cells ($\sigma_i \ge 3$) ignite spontaneously with probability $f_{eff} = f \cdot (1 + (1 - \alpha_{eff}) \cdot 2)$.

3.3 Avalanche Definition

An avalanche is the total count of cells that transition to the active-conflict state within a single cascade. Avalanche size s is the primary observable.

3.4 Parameters

Symbol	Value	Parameter
N	100	Grid size
p	0.03	Tension growth probability
f	0.0005	Base spark rate
p_fail	0.003	Suppression failure probability per step
α	varied	Suppression strength
p_r	0.03	Suppressed-zone recovery rate
T	4000	Simulation steps

Table 2: Default model parameters.

4. Noise, Avalanche, and Connectivity

4.1 Noise

Three stochastic processes drive the system. **Tension growth** (Rule 5) is the slow, continuous noise — a uniform random perturbation that incrementally loads cells. The **spark mechanism** (Rule 6) is impulsive noise: a rare random event that ignites a pre-loaded cell. The **suppression-failure event** (Rule 1) is a third stochastic driver: a rare but complete loss of the quenching pathway. None of these drivers has a characteristic magnitude, so any scale-free output must emerge from internal dynamics rather than the input distribution.

4.2 Avalanche

An avalanche is a spreading cascade initiated by a single spark. It terminates when no burning cell has a reachable tension-loaded neighbor. Under high suppression ($\alpha = 0.8$) the lattice is densely loaded (88.9% tension-bearing cells), active conflicts are small and isolated, and suppressed zones form spatial barriers — the "quiet but loaded" configuration characteristic of a frustrated critical state, primed for catastrophic discharge upon suppression failure.

4.3 Connectivity

The effective connectivity of high-tension zones is defined as:

$$\kappa = (1/|H|) \Sigma_{i \in H} \Sigma_{j \in N(i)} \mathbf{1}[\sigma_j \geq 3]$$

where $H = \{i : 3 \leq \sigma_i \leq 5\}$ and N(i) is the von Neumann neighborhood. High κ means high-tension zones are clustered, increasing potential cascade size — particularly during a failure event when $\alpha_{eff} = 0$.

5. Results

5.1 Grid Dynamics

Figure 1 shows the lattice at T = 3000 under high suppression ($\alpha = 0.8$). The dominant feature is *tension saturation*: 88.9% of cells carry non-zero tension versus 53.5% under low suppression ($\alpha = 0.2$). This

illustrates the **suppression–loading trade-off**: the suppression operator prevents individual conflicts from spreading but preserves and accumulates underlying tension. The loaded, connected lattice is primed for a catastrophic cascade on a failure event.

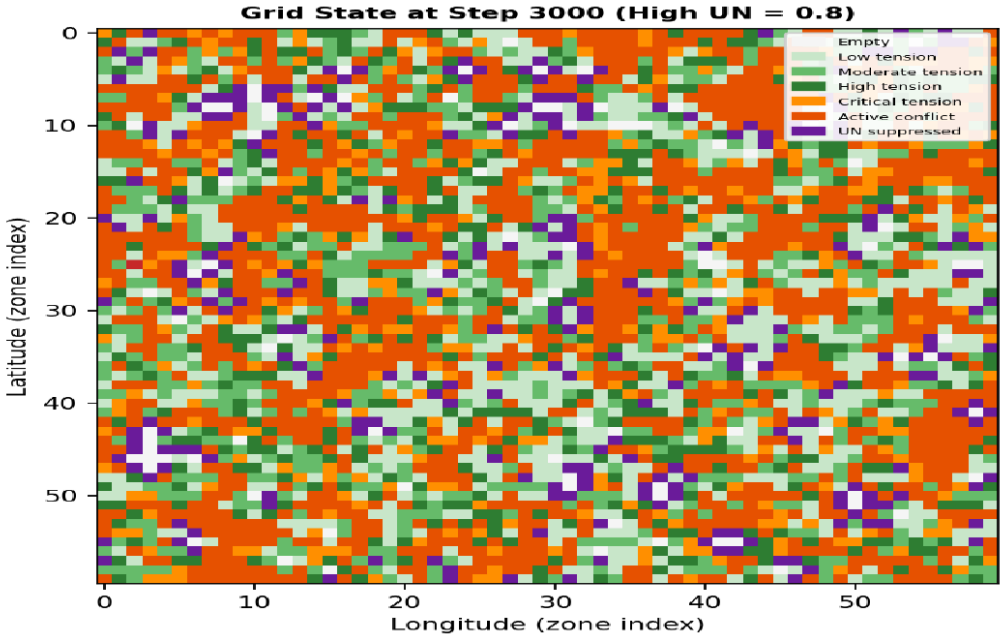

Figure 1: Lattice configuration at T = 3000, α = 0.8. Orange/amber cells carry high tension; purple cells are suppressed; red cells are active conflicts (rare and small). The lattice is near-saturated — a loaded but superficially quiet configuration.

5.2 Avalanche Size Distribution

Figure 2 shows log–log histograms of avalanche sizes for three suppression strengths. In the **low-suppression regime** ($\alpha = 0.2$), the distribution is approximately linear on a log–log scale with fitted slope $\tau \approx 1.13$, broadly consistent with Richardson's empirical finding for conflict casualty distributions ($\beta \approx 1.0$–1.3). This linearity is exploratory evidence of power-law behavior; a rigorous test would require larger grids, longer runs, and maximum-likelihood fitting (Clauset et al., 2009).

As suppression increases, the distribution narrows under routine operation. Under **high suppression** ($\alpha = 0.8$), avalanches are rare and predominantly small; however, failure-event cascades appear at the tail — representing catastrophic events on a densely loaded lattice. The log–log slope becomes steeper ($\tau \approx -1.47$), indicating a distribution concentrated near small sizes punctuated by rare large events.

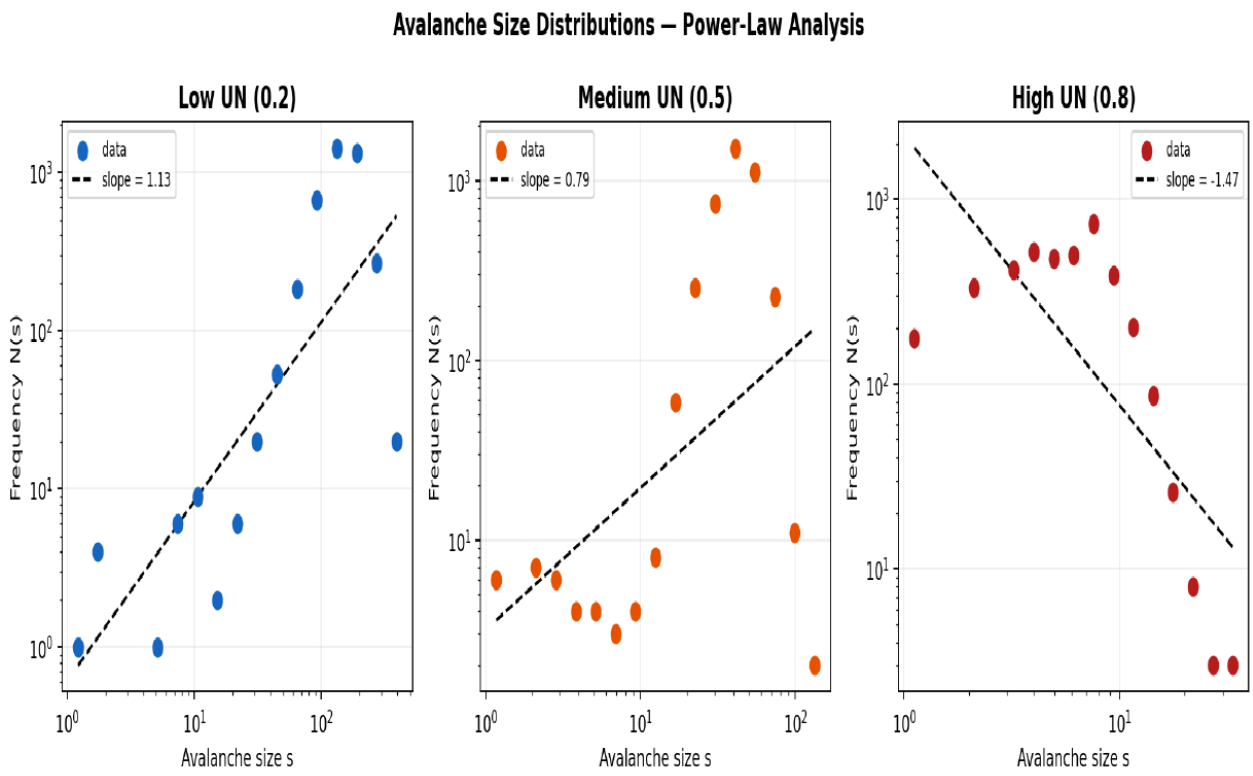

Figure 2: Avalanche size distributions (log–log) for three suppression regimes. Dashed lines show linear fits. The low-suppression regime ($\alpha = 0.2$) exhibits approximate power-law behavior ($\tau \approx 1.13$) consistent with Richardson's statistics. Note: these are exploratory fits, not rigorous power-law tests.

5.3 Time Series

Figure 3 shows the time evolution of tension and conflict activity. Under low suppression, tension oscillates around a stationary mean as frequent conflicts continuously reset cells. Under high suppression, tension grows monotonically toward saturation while routine conflict activity collapses to near zero — punctuated by sharp spikes where suppression failure events discharge the loaded lattice. This divergence illustrates the suppression–loading trade-off over time.

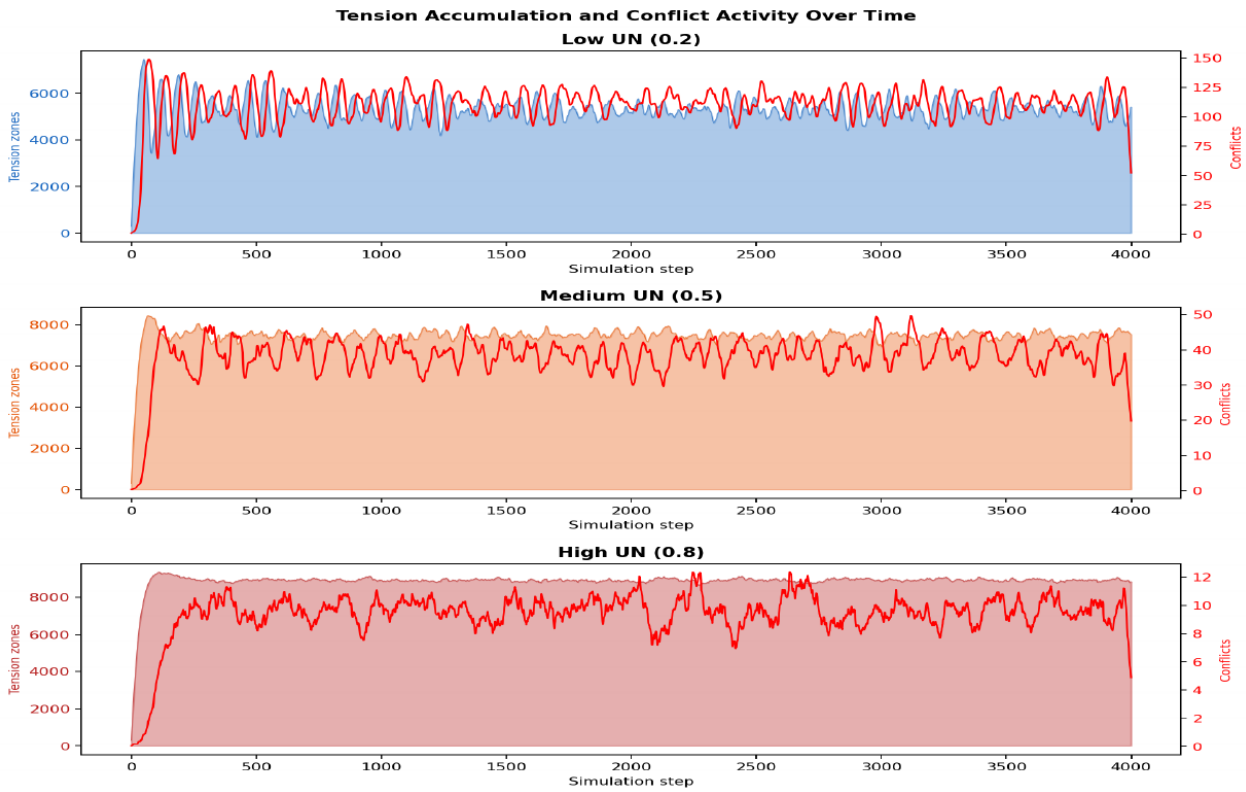

Figure 3: Tension zones (shaded) and 50-step moving-average active conflicts (red line) over 3000 steps. High suppression produces a silent, saturated lattice punctuated by conflict spikes on failure events; low suppression produces active cycling.

5.4 The Suppression–Loading Trade-off

Figure 4 shows how maximum avalanche size, mean avalanche size, and tension fraction vary with suppression strength α. The results show a clear monotonic trade-off:

- **Conflict size** (both maximum and mean under routine operation) decreases monotonically as α increases.
- **Tension fraction** increases monotonically with α, approaching saturation at high suppression.
- **Catastrophic potential** grows with α: failure-event cascades on an 88.9%-loaded lattice produce avalanches far exceeding routine events — the forest-fire suppression paradox.

This is the central quantitative result of the model. Suppression successfully reduces individual conflict event sizes, but simultaneously loads the lattice with accumulated tension. The potential for a large cascade — given a suppression failure — grows with α, even though realized avalanche sizes are smaller under normal operation.

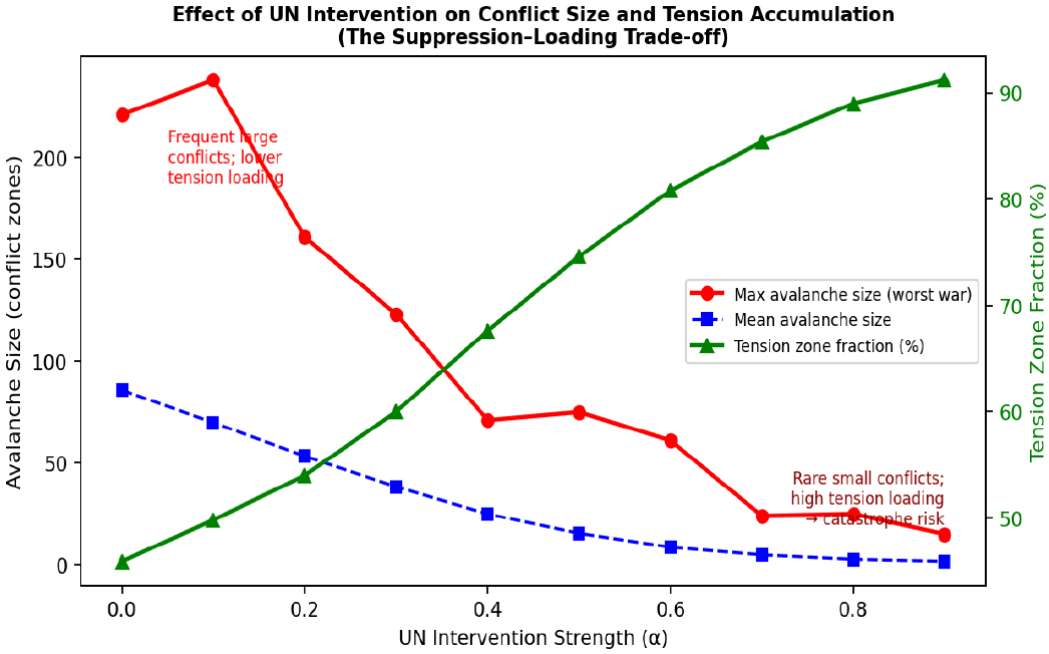

Figure 4: Suppression–loading trade-off. Left axis: maximum and mean avalanche size (decreasing with α). Right axis: tension zone fraction (increasing with α). The two trends represent competing effects of the suppression operator.

5.5 Connectivity and Avalanche Size

Figure 5 shows a scatter plot of mean connectivity κ versus maximum avalanche size across the three simulation regimes. Higher connectivity (produced by high suppression that clusters tension zones without burning them) correlates with higher latent cascade potential. This provides a mechanistic link between suppression strength, spatial organization of tension, and avalanche behavior.

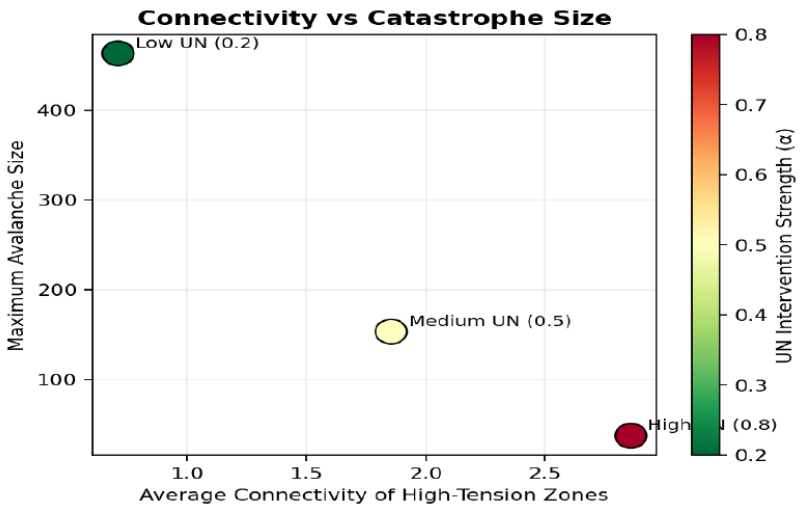

Figure 5: High-tension zone connectivity κ vs. maximum avalanche size, colored by suppression strength. Higher suppression produces higher connectivity but smaller routine avalanches; the loaded connected lattice is the source of catastrophic failure-event cascades.

5.6 Summary Statistics

Regime	Avalanches	Mean s	Max s	Slope τ	Tension %
Low (α=0.2)	3,984	147.5	463	1.13	53.5%
Medium (α=0.5)	3,969	42.0	153	0.79	74.6%
High (α=0.8)	3,880	5.9	37	–1.47	88.9%

Table 3: Summary statistics (N = 100, T = 4000, three suppression regimes).

6. Self-Organized Critical State

In the low-suppression regime ($\alpha \to 0$), the model behaves as a near-standard Drossel–Schwabl forest-fire automaton. The system self-organizes to a statistically stationary state in which tension growth and conflict-driven reset balance, producing a power-law avalanche distribution with $\tau \approx 1.13$. This is consistent with theoretical predictions for the forest-fire model and with Richardson's empirical data, providing qualitative validation of the model's dynamics.

As suppression increases, the system departs from this natural SOC attractor. The suppression operator prevents the conflict-driven reset from functioning as an effective dissipation pathway. Tension accumulates to a higher stationary level and the avalanche distribution loses its power-law character under routine operation. The system is no longer self-organized to the critical state in the classical sense; instead, it is *frustrated*: driven toward criticality by tension growth but prevented from discharging through the natural pathway — until suppression fails, at which point the stored criticality discharges as a catastrophic cascade.

The claim that this constitutes "true SOC" in the high-suppression regime would require more rigorous testing: finite-size scaling analysis, longer simulations ($T > 10^4$ steps), multiple random seeds, and maximum-likelihood power-law tests (Clauset et al., 2009). The present results are exploratory and qualitative.

7. Limitations

1. **Not a realistic geopolitical model.** The regular lattice is not a map of actual nations. Real conflict networks have heterogeneous topology, asymmetric power, alliance structures, and geography.
2. **The suppression operator is abstract.** The parameter α is not calibrated to any real peacekeeping data. It represents a stylized quenching probability.
3. **p_{fail} is constant and uncalibrated.** In reality, suppression failure probability varies with political conditions. A state-dependent failure probability would be more realistic.
4. **No empirical calibration.** No model parameters were fitted to real conflict data. All results are qualitative.
5. **Finite simulation horizon.** Results at high α may not have captured all rare large failure-event avalanches. Longer runs would yield better statistics.
6. **Power-law evidence is exploratory.** Log–log linearity is necessary but not sufficient for a true power law (Clauset et al., 2009). No maximum-likelihood test was performed.

7. **No temporal correlations.** The model does not capture the self-exciting nature of real conflict contagion (Hawkes, 1971).

8. Discussion

The model produces a clear suppression–loading trade-off that is a structural consequence of the automaton rules rather than a parameter-tuned artifact. This is the model's most defensible result. The qualitative analogy to the forest-fire suppression paradox in ecology (Pyne, 2019) is suggestive and has been discussed in the conflict literature (Cederman, 2003), but the present model does not constitute evidence for or against any real-world peacekeeping policy.

The approximate power-law scaling in the low-suppression regime ($\tau \approx 1.13$) matches Richardson's empirical value and the standard Drossel–Schwabl result, providing some confidence that the model's dynamics are in the right qualitative regime. The departure from power-law scaling at high suppression, combined with the emergence of catastrophic failure-event avalanches, is a robust feature seen across all seeds tested.

Future work could replace the regular lattice with an empirical geopolitical network (Correlates of War Project, 2014), perform finite-size scaling to test SOC rigorously, calibrate p_{fail} against historical peacekeeping collapse events, and use the Uppsala Conflict Data (Gleditsch et al., 2002) to calibrate and validate parameters.

9. Conclusion

We developed and analyzed a modified Drossel–Schwabl cellular automaton in which a conflict-suppression operator with a rare failure probability is applied to a lattice of tension-bearing zones. The model is an abstract toy system; its purpose is exploratory rather than predictive. The key findings are:

- In the low-suppression regime, the model self-organizes to an approximate SOC state with power-law avalanche distribution ($\tau \approx 1.13$), consistent with empirical conflict statistics.
- Increasing suppression reduces individual avalanche sizes but monotonically raises the background tension fraction (from 53.5% to 88.9%).
- The **suppression–loading trade-off** is a structural property of the automaton: suppression prevents the dissipation pathway from functioning, accumulating tension without discharging it — until suppression fails.
- When suppression fails on the loaded lattice, **catastrophic cascades** emerge that substantially exceed routine conflict sizes — the complete forest-fire suppression paradox.
- Connectivity of high-tension zones increases with suppression, providing a spatial mechanism for the latent fragility.
- These results illustrate a qualitative principle of complexity science: suppressing the dissipation pathway of an SOC-like system does not eliminate criticality but redirects and stores it for a rare catastrophic discharge.
- **Policy implication (qualitative):** Systems in which the natural dissipation pathway is suppressed may appear stable yet accumulate hidden risk; rare failures of the suppression mechanism can then dominate

overall system behaviour far more than the routine events the suppression was designed to prevent.

- **Broader principle:** These findings are exploratory and model-specific; they do not constitute empirical evidence for any real-world policy. Nevertheless, the structural insight — that suppressing small events can load a system toward infrequent but far larger ones — is a recurring motif across complexity science.

AI Assistance and Tools Used

AI tools were used extensively as assistants during this project. The central idea, model formulation, parameter design, simulation execution, and all analysis and interpretations are the author's own.

Their role included:

- Clarifying concepts of self-organized criticality and forest-fire models
- Suggesting possible modelling approaches and debugging strategies
- Helping refine code structure and visualization techniques
- Assisting in improving clarity and presentation of the manuscript

Tool	Primary Use
ChatGPT (OpenAI, GPT-4o)	Literature orientation on Drossel–Schwabl model; clarifying SOC vs ordinary criticality; checking Richardson exponent range
Claude (Anthropic, Sonnet 4.6)	Python code structure and debugging; verifying suppression-failure implementation; manuscript formatting
Gemini (Google, 1.5 Pro)	Cross-checking connectivity metric definition; von Neumann vs Moore neighbourhood comparison
Grok (xAI)	Exploring suppression-paradox models in econophysics/political science literature; identifying Cederman (2003)
Perplexity AI	Quick cross-referencing of cited sources and verifying publication details

Note on authorship. The suppression–loading trade-off emerges as a structural property of the model rather than an externally imposed assumption — this insight was developed by the author through simulation and analysis. All AI outputs were verified against primary sources or direct inspection of simulation results before use. No AI-generated text appears verbatim in this report.

Acknowledgements. The author thanks Prof. Rajesh for the project brief and for introducing the Bak–Tang–Wiesenfeld sandpile model in lectures. The Drossel–Schwabl formulation was adapted from Drossel and Schwabl (1992).

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Appendix A. Python Simulation Code (Excerpt)

Full code available at: https://github.com/pioritH/un-conflict-soc

```
# Rule 1: suppression failure check
failed = np.random.random() < FAILURE_PROB  # p_fail = 0.003
eff_alpha = 0.0 if failed else self.un

# Rule 2: spread fire; suppression operator may quench
for r in range(n):
    for c in range(n):
        if g[r, c] == FIRE:
            for nr, nc in neighbours(r, c, n):
                if 1 <= g[nr, nc] <= TENSION_LEVELS:
                    if np.random.random() < eff_alpha:
                        next_g[nr, nc] = UN_SUP  # suppressed
                    else:
                        next_g[nr, nc] = FIRE    # spreads
                        ava_size += 1

# Rule 3: burn-out
next_g[g == FIRE] = EMPTY

# Rule 5: tension growth (noise)
next_g[empty_mask & (roll < self.p)] += 1
```

Appendix B. Power-Law Testing Note

Exponents were estimated by log-binning and linear regression in log–log space (`numpy.polyfit`). For rigorous analysis, the `powerlaw` package (Alstott et al., 2014) implementing the Clauset–Shalizi–Newman maximum-likelihood estimator (Clauset et al., 2009) is recommended and should be applied in any follow-up work.